Unique Electrodes™

Down to Earth....

Complete

Earthing & Lightning

Protection Solutions

Manufacturer of
Chemical Erthing & Lighting
Protection System



TESTED BY

CPRI APPROVED

ISO 9001:2015

NABL

ERDA

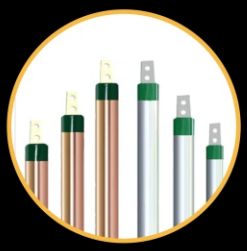
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ABOUT UNIQUE ELECTRODES

Unique Electrodes is committed to delivering high-quality earthing and lightning protection solutions for industrial, commercial, infrastructure, and residential applications.

With a focus on quality, innovation, and safety, we provide products that help protect people, equipment, and electrical systems from faults and lightning-related risks.

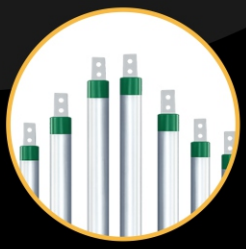
WHAT WE OFFER



Chemical Earthing Electrodes



Copper Bonded Earthing Rods



Hot Dip Galvanized Earthing



Earth Enhancement Compound



Funnel Earthing



Lightning Protection Systems



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OUR PRODUCT

COPPER-COATED PIPE TYPE EARTHING ELECTRODES



Our Copper-Coated Pipe Type Earthing Electrodes are engineered to provide efficient, safe, and long-lasting grounding solutions for a wide range of electrical installations. Manufactured using high-quality steel pipes with a uniform copper coating, these electrodes offer excellent electrical conductivity, superior corrosion resistance, and reliable performance even in challenging soil conditions.

Available in both Pipe-in-Pipe and Flat-in-Pipe configurations, the electrodes are filled with highly conductive Conductive Carbon Compound (CCC), which helps reduce earth resistance and improves fault current dissipation. This advanced design ensures enhanced grounding efficiency, increased current-carrying capacity, and stable performance throughout their service life.

HOT-DIP GALVANIZED PIPE TYPE EARTHING ELECTRODES



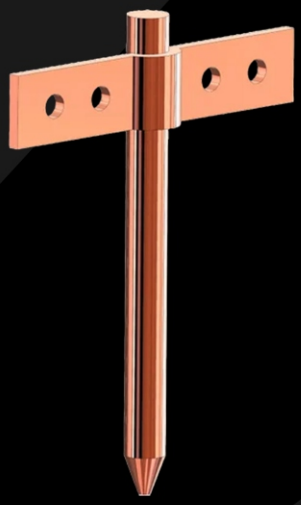
Our Hot-dip Galvanized Pipe Type Earthing Electrodes are manufactured using premium-quality steel pipes with a durable zinc coating for superior corrosion protection and long service life. Designed to provide efficient grounding and low earth resistance, they ensure reliable fault current dissipation and enhanced electrical safety.

Available in 48mm and 76mm diameters, these electrodes are ideal for residential, commercial, industrial, telecom, and lightning protection applications. Their robust construction and maintenance-free performance make them a dependable choice for long-lasting earthing systems.

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COPPER BONDED SOLID ROD

Engineered for durability and efficiency, Copper Bonded Solid Rods offer outstanding conductivity and long-term protection against corrosion. Their solid-core design ensures high mechanical strength, making them suitable for demanding grounding environments and deep-driven installations.



Available in 14mm, 17mm, 25mm, 32mm, and 38mm sizes, these rods provide a dependable and cost-effective earthing solution for industrial, commercial,and utility applications.

EARTH ENHANCEMENT COMPOUND

The efficiency of an earthing system is greatly influenced by the soil conditions surrounding the earthing electrode. Even when the same electrode and installation method are used, variations in soil composition can lead to significant differences in earth resistance and overall system performance.

Earth Enhancement Compound (EEC) is specially formulated to improve the conductivity of the soil around the earthing electrode. When mixed with the backfill soil during installation, it reduces soil resistivity and creates a stable, low-resistance path for fault current, ensuring reliable and consistent earthing performance.



STANDARD BFC BAG



STAR BFC BAG

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LIGHTNING ARRESTER



Single Spike
Lightning Arrester

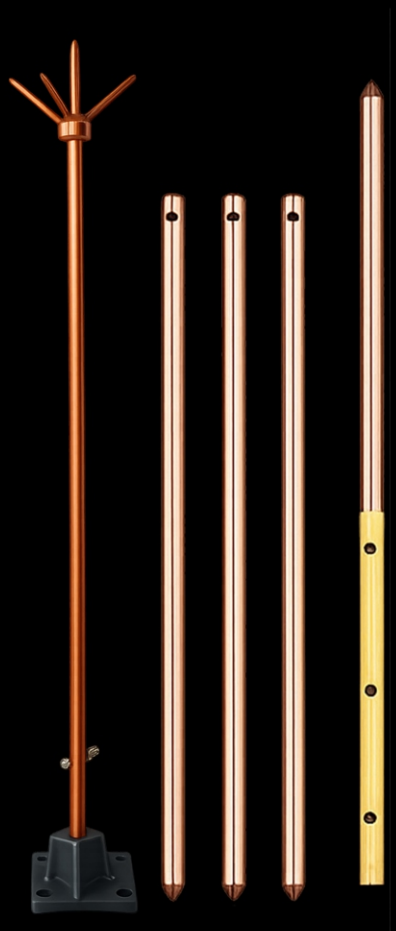


Multi Spike
Lightning Arrester



ESE Type
lightning arrester

SOLAR EARTHING KIT



Our Solar Earthing Kit is specially designed for rooftop and ground-mounted solar power projects. The kit consists of a high-strength copper bonded steel earthing rod available in 14mm and 17mm diameters with copper bonding thickness ranging from 50 Micron to 250 Micron. It provides excellent corrosion resistance, low earth resistance, and long service life, ensuring reliable protection for solar panels, inverters, structures, and electrical equipment.

We also offer Single Spike Lightning Arrester and Multi Spike Lightning Arrester solutions for complete lightning protection of solar power plants. These systems are designed to safely intercept lightning strikes and discharge the surge current to the ground through a reliable earthing network, helping protect solar modules, inverters, electrical panels, and other critical equipment from lightning-related damage.

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PRODUCT DIMENSIONS & TECHNICAL SPECIFICATIONS

EARTHING TYPE	COATING	LENGTH	MICRON	OUTER ELECTRODE DIA	INNER STRIP SIZE
Copper Coated Earthing	Copper Coating	1.2,3 Mtr	50 Micron 100 Micron 250 Micron	48,60,76,88 mm	Strip: 25x3, 25x6, 32x6, 40x6, 50x6, 50x10
Hot-dip Galvanized Earthing	Hot-dip Galvanized	1.2,3 Mtr	80 Micron	48,60,76,88 mm	Strip: 25x3, 25x6, 32x6, 40x6, 50x6, 50x10
Solid Rod Earthing	Copper Coating	1.2,3 Mtr	50 Micron 100 Micron 250 Micron	14,17,19,25,32,38 mm	N.A.

INSTALLATION METHOD

It is very important that the earthing system is installed correctly for the system to work at its full efficiency. upon proper installation, the system will give very good (desired resistance value) compare to conventional system of earthing. It will require no maintenance under normal soil conditions except the normal watering of pits during hot summer seasons. The standerd installation method description is packed with every electrode manufactured by us.

NOTE: Since there exist varying types of soil& environment conditions around the best resistance values of an earthing system vary.

NORMAL SOIL

- Make a bore of 8" to 10" in dia up to the electrode length of 2/3 meters.
- Fill the bottom 4" of the bore with the supplied BFC (Electrodite).
- Vertically place the electrode in the center of the pit.
- Fill the cavity around the electrode with the BFC (Electrodite).
- Ensure there are no air gaps in the BFC Filled cavity region.
- Now pour sufficient water into the earth pit unit the BFC takes the from of a paste / mud. allow the pit to absorb the water and settle.
- Test the earth pit and make the required connections to the electrical service box.

CAUTION: avoid excess watering and don't hammer the electrode.

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RECOMMENDATION OF INSTALLATION OF EARTHING AS PER IS:3043

INSTALLATIONS/3 PHASE SHORT-CIRCUIT CURRENT (ISC CAPACITY)	IR VALUE REQUIRED	SOIL TYPE/RESISTIVITY	EARTH SYSTEM
House hold earthing / 3kA	8 ohm	Normal Soil / up to 50 ohm-meter	Single Electrode
		Sandy Soil / between 50 to 2000 ohm-meter	Single Electrode
		Rocky Soil / more than 2000 ohm-meter	Multiple Electrodes
Commercial premises, Office /5kA	2 ohm	Normal Soil / up to 50 ohm-meter	Single Electrode
		Sandy Soil / between 50 to 2000 ohm-meter	Multiple Electrodes
		Rocky Soil / more than 2000 ohm-meter	Multiple Electrodes
Transformers, substation earthing, LT line equipment /15kA	less than 1 ohm	Normal Soil / up to 50 ohm-meter	Single Electrode
		Sandy Soil / between 50 to 2000 ohm-meter	Multiple Electrodes
		Rocky Soil / more than 2000 ohm-meter	Multiple Electrodes
LA, High current equipment / 50kA	less than 1 ohm	Normal Soil / up to 50 ohm-meter	Single Electrode
		Sandy Soil / between 50 to 2000 ohm-meter	Multiple Electrodes
		Rocky Soil / more than 2000 ohm-meter	Multiple Electrodes
PRS, UTS, RTUS, Data processing center etc./5kA	less than 0.5 ohm	Normal Soil / up to 50 ohm-meter	Single Electrode
		Sandy Soil / between 50 to 2000 ohm-meter	Multiple Electrodes
		Rocky Soil / more than 2000 ohm-meter	Multiple Electrodes

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OTHER PRODUCTS & ACCESSORIES



CAST IRON EARTHING



COPPER STRIP



GI STRIP



RCC CHAMBER



PPE CHAMBER

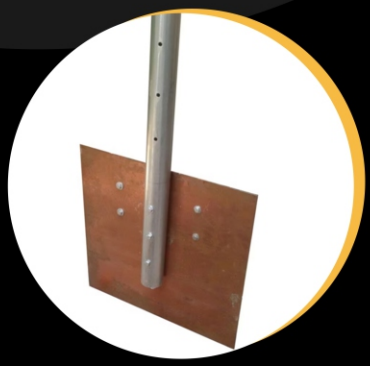
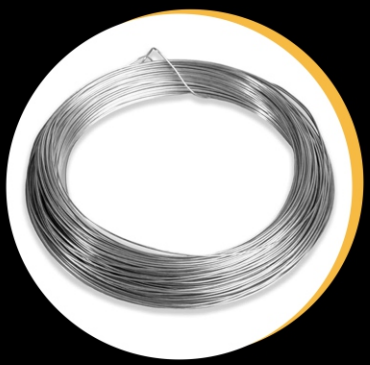


PLATE EARTHING



GI WIRE



EARTHING CLAMP



LIGHTING ARRESTER ACCESSORIES

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APPLICATION AREAS



RENEWABLE ENERGY



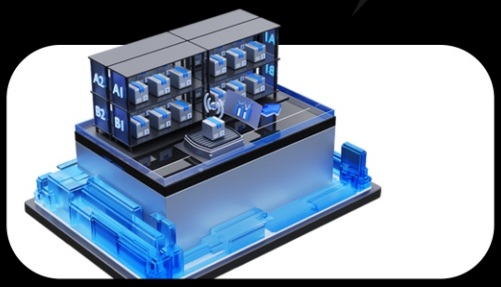
INDUSTRIAL FACILITIES



POWER & ENERGY



MANUFACTURING PLANTS



DATA CENTERS



GOVERNMENT PROJECTS



RAILWAY



RESIDENTIAL & COMMERCIAL PROJECTS



ELECTRICAL SUBSTATION

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